©VIT IPR & TTCELL

(insert VIT logo) **Invention Disclosure Format (IDF)-B**	Document No.	
	Issue No/Date	
	Amd. No/Date	0/00.00.0000

1. Title of the invention:

"Hardware-Attested Continuous Security-Risk Scoring of Cloud Infrastructure Resources Using Multi-Rate Platform Attestation and Baseboard Sensor Telemetry"

2. Field / Area of invention

- This invention relates to the field of cloud computing infrastructure security, specifically to cyber-physical measurement apparatus that integrate hardware-rooted platform attestation evidence with baseboard environmental sensing and processor event counters to quantify the security exposure of provisioned cloud resources.
- The invention pertains to an apparatus and method that acquires Trusted Platform Module attestation quotes, baseboard management controller sensor readings, processor performance-monitoring-unit counters and cloud provider configuration records at differing sampling rates, fuses them under a per-signal staleness weighting into a composite resource-level risk score accompanied by a confidence attribute, and applies threshold crossings of that score directly to enforcement hardware comprising network interface controller filter tables and host power-delivery registers.
- The scope of this invention explicitly excludes configuration-only cloud security posture scanners, vulnerability catalogue scoring schemes operating on published severity metrics, graph analytics over declarative resource relationships without physical measurement, binary platform attestation verdicts that produce no graded score, and any arrangement that reports risk to an operator without actuating hardware in the measured infrastructure.

3. Prior patents and publications from literature

The following representative prior art was reviewed to map gaps relevant to the present invention:

Patent / Publication / Year	Core Contribution	Data / Method Focus	Limitations / Gaps Identified
US 2024/0354213 A1 (Patent, 2024) Palo Alto Networks, Inc.	Graph representation of cloud resources and security-vendor-defined assets, analysed to rank which resources an incident would affect.	Configuration data drawn from provider APIs; PageRank, strongly-connected-component and betweenness centrality over a directed resource graph.	Criticality is inferred wholly from declarative configuration relationships. No measurement of host firmware or physical operating state is taken; output is an analytical report rather than a signal that actuates enforcement hardware.
US 12,316,667 B2 (Patent, 2025)	Supervised machine-learning risk scoring applied to cloud permission assignments.	Labelled entitlement records from the cloud control plane with classical supervised classifiers.	Scores a single control-plane artefact class. Contains no hardware-rooted measurement, no treatment of signal staleness, and no

			coupling between the produced score and any actuation path.
US 12,495,061 (Patent, 2025)	Agentless posture scanning that correlates misconfigurations, vulnerabilities and identity activity, then ranks remediations by return on investment.	Machine-learning threat correlation across scan findings from multiple providers.	Correlation operates over findings gathered on a single slow scan cadence, so signals of differing sampling periods cannot be reconciled. The system recommends action to an operator rather than applying graded actuation to hardware.
US 12,542,801 (Patent, 2025)	Dynamic prioritisation of attack paths in cloud-native applications over an asset graph.	Scanner-built asset graph augmented from external datastores; software bill-of-materials and connectivity metadata.	Path scoring rests on software inventory and topology. Per-host integrity evidence is absent, as is any physical sensing or staleness-aware weighting.
US 11,768,945 B2 (Patent, 2023)	Prediction of revised temporal vulnerability scores, adjusted by organisational mitigating factors into an enterprise risk score.	Machine-learning regression over published vulnerability catalogues and organisational policy records.	Operates on catalogue and policy data and is decoupled from the runtime condition of any particular host machine; cannot respond to firmware substitution or anomalous physical behaviour.
US 11,886,610 B2 (Patent, 2024) Indian priority 202311035748	Cloud database log analyser detecting risk signatures for posture and data-security management.	Signature detection over cloud database and audit log streams.	Log-derived signatures are software-plane artefacts. A host whose measured boot chain has been altered can continue to emit entirely clean log records, leaving the compromise unscored.
US 12,321,491 (Patent, 2025) Indian priority 202311050873	Risk event detection combining account activity with graph-based context for cloud data security posture management.	Activity streams joined to a resource relationship graph.	Both inputs originate in the control and data planes. Platform attestation evidence plays no part, and no physical enforcement follows from a detected event.
US 11,917,083 B2 (Patent, 2024) VMware, Inc.	Host attestation performed through a smart network interface controller acting as an attestation intermediary with reduced attack surface.	Two-phase attestation protocol between the controller, the host and a security authority.	Yields a binary trusted or untrusted verdict at attestation time. No graded, continuously updated resource-level score is derived, and attestation evidence is not fused with sensor or configuration data.
EP 3217310 A1 (Patent, 2017)	Hypervisor-based attestation of virtual environments through virtual platform module credentials bound to physical register values.	Trusted-computing attestation protocol over virtualised platforms.	Provides a point-in-time integrity check only. There is no scoring construct, no fusion across measurement rates, and no downstream actuation of enforcement hardware.
US 11,221,940 B2 / IN 201811031336 (Patent, 2022) Accenture Global Solutions	Machine-learning selection of virtual computing node types for an application from measured performance parameters.	Supervised models over historical node performance measurements.	Directed at performance and cost of node selection. Security exposure of the selected host is neither modelled nor measured.
Mell, Scarfone & Romanosky, IEEE Security & Privacy (Journal, 2006)	Defines the Common Vulnerability Scoring System, the base severity framework underlying most contemporary risk ranking.	Expert-defined metric groups producing a static base severity per vulnerability.	Severity is static and context-free. The score describes a vulnerability class rather than the measured condition of a specific deployed resource on a specific host.

Jacobs et al., Digital Threats: Research and Practice (Journal, 2021)	Introduces the Exploit Prediction Scoring System, estimating the probability that a vulnerability will be exploited in the wild.	Logistic modelling over large-scale exploitation telemetry and vulnerability metadata.	Produces a population-level probability per vulnerability identifier. It cannot express the exposure of one cloud resource and consumes no platform measurement.
Jung, Li & Bechor, Computers & Security, vol. 116 (Journal, 2022)	Context-aware vulnerability prioritisation model adjusting severity by environmental context.	Context features drawn from asset inventory and organisational records.	Context originates in administrator-maintained inventories rather than measured hardware state, so stale inventory silently corrupts the ranking with no confidence indication.
Wang, Neil & Fenton, Computers & Security, vol. 89 (Journal, 2020)	Bayesian network implementation of the FAIR model for quantifying cyber risk at organisational level.	Expert-elicited prior distributions propagated through a probabilistic graphical model.	Requires periodic expert reassessment and is not a continuous measurement apparatus; provides no interface to enforcement hardware and no per-resource resolution.
Zhao et al., Computers & Security (Journal, 2023)	Survey of techniques for constructing cybersecurity knowledge graphs.	Literature review of extraction, fusion and reasoning methods over textual and threat-intelligence sources.	Graphs are populated from documentary sources. Sensor telemetry and hardware attestation evidence fall entirely outside the surveyed construction methods.
Nour et al., SPGNN-API (Journal, 2023)	Graph neural network identifying attack paths and tuning zero-trust network policy to break high-risk paths.	Inductive GNN over network topology with vulnerability severity metrics as node features.	Node features derive from severity metrics and topology. Enforcement is confined to network policy configuration and is not anchored in attested platform measurement or physical sensing.
Wang et al., CloudShield (Journal, 2022)	Real-time anomaly detection for cloud hosts from hardware performance counter reconstruction error.	Unsupervised reconstruction models over processor performance-monitoring-unit event streams.	Reports a host-level anomaly indication against a threshold. The indication is not attributed to individual cloud resources, is not fused with attestation or configuration context, and drives no enforcement channel.
Hosseinzadeh et al., Security and Privacy (Wiley), vol. 3 (Journal, 2020)	Reviews application of Trusted Platform Module functionality to cloud computing, covering trusted boot, remote attestation and sealed storage.	Survey of trusted-computing mechanisms and their cloud deployment patterns.	Catalogues attestation mechanisms without proposing any quantification framework converting attestation evidence into a graded, time-varying risk measure with an explicit confidence attribute.
Hore, Shah & Bastian, Expert Systems with Applications, vol. 221 (Journal, 2023)	Deep reinforcement learning for sequential allocation of vulnerability remediation resources under budget constraints.	Reinforcement learning agent consuming vulnerability scores and organisational constraints.	Consumes risk scores as given inputs. The work presupposes an upstream scoring mechanism and does not address how such a score is measured at the physical platform.

4. Summary and background of the invention

Cloud estates now span thousands of short-lived compute, storage and network resources whose security condition is assessed almost entirely by reading configuration metadata through provider application programming interfaces. Metadata of this kind is declarative: it records how a resource was intended to be configured and remains silent on whether the firmware, measured boot chain or physical

operating envelope of the underlying host has been altered. A machine whose boot chain has been substituted continues to report a clean posture until some downstream symptom surfaces in a later scan cycle, and published analyses of cloud posture tooling consistently identify this blindness to platform state as a dominant failure mode.

A second deficiency compounds the first. Existing scoring arrangements treat every input as though it arrived on one cadence, so a configuration record retrieved five minutes ago carries the same weight as a processor counter read a tenth of a second ago. Sampling periods across the signals relevant to cloud risk span roughly five orders of magnitude, and collapsing them into a single unweighted vector understates exposure during precisely the interval in which a compromise is most likely to remain undetected. The resulting figure also carries no expression of its own reliability, so a consuming system cannot distinguish a score computed from complete evidence from one computed while a sensor channel was unavailable.

The critical gap lies in the absence of an apparatus that concurrently (i) acquires signed attestation evidence from a hardware root of trust seated on the host mainboard together with baseboard environmental readings and processor event counters, (ii) reconciles these streams against slower configuration records under a per-signal staleness decay matched to each acquisition period, (iii) emits a bounded composite score accompanied by a confidence attribute derived from realised weight retention, and (iv) applies threshold crossings of that score directly to enforcement hardware in the measured infrastructure. The present invention discloses an apparatus and method composing these four functions into a single integrated measurement and enforcement pipeline, thereby converting periodic advisory reporting over administrative records into continuous physical measurement that acts on the infrastructure it measures.

The present invention is architecturally and functionally distinct from all cited prior art references. No cited system combines hardware-rooted attestation evidence, baseboard environmental sensing, processor event counters and configuration context within one staleness-weighted composite score that gates physical actuators. It does not assert or depend on:

- Configuration-only posture scanners that read provider metadata without measuring platform state;
- Vulnerability catalogue scoring schemes operating on published severity or exploitation metrics;
- Graph analytics over declarative resource relationships without any physical measurement input;
- Binary platform attestation protocols that return a trusted or untrusted verdict without a graded score; or
- Host anomaly detectors that raise a threshold alarm without attribution to individual cloud resources or coupling to enforcement hardware.

Instead, the invention is confined to a unique attestation-anchored, staleness-weighted, score-gated workflow that produces confidence-annotated resource-level risk scores not previously claimed, functions within strictly defined sampling and decay parameters, and generates outputs solely for the technical purpose of continuous cloud infrastructure risk measurement and physical enforcement. This intentional architectural limitation guarantees non-overlap, promotes originality, and establishes a distinct inventive step beyond both the existing prior art and any related published frameworks.

5. Objective(s) of the invention

- To provide a cyber-physical measurement apparatus that derives a continuously updated, bounded security-risk score for each cloud resource from Trusted Platform Module attestation quotes,

baseboard management controller sensor readings and processor performance-monitoring-unit counters acquired from the physical host on which that resource executes, thereby resolving compromise indications that configuration metadata cannot express.

- To enable fusion of measurement streams whose acquisition periods span approximately five orders of magnitude into a single calibrated score through a per-signal exponential staleness decay, and to emit alongside the score a confidence attribute computed as realised weight retention, such that a consuming system can distinguish evidential completeness from evidential absence.

- To validate a score-gated enforcement subsystem that writes filter entries into the ternary content-addressable memory of the host network interface controller and adjusts running average power limit registers of the processor package upon defined threshold crossings, delivering containment that takes effect in the hardware datapath before packets reach host memory and without operator intervention.

5A. Novelty and Key Advantage of the Invention

The present invention establishes a dedicated attestation-anchored, staleness-weighted, score-gated risk measurement apparatus and enforcement workflow that is architecturally distinct from all previously developed cloud security posture and vulnerability scoring systems.

Unlike prior systems that concentrate on configuration-only posture scanning, catalogue-based severity scoring, graph analytics over declarative relationships, or binary platform attestation, this invention is limited to a specifically defined six-zone pipeline carried out through an independent, hardware-rooted measurement workflow. The attestation collector and integrity delta evaluator, the multi-rate buffer and staleness weighting unit, the composite scoring engine with its confidence estimator, and the gating controller with its two physical actuation channels are all specifically designed for and confined to this invention.

A significant benefit of this invention is its clear architectural separation:

- It does not reuse, extend, or rely on any workflows, system components, data pipelines, or analytical outputs claimed in earlier posture-management, vulnerability-scoring, or attestation patents.

- It prevents functional overlap by operating on distinct hardware-rooted measurement inputs, independent sideband acquisition channels, and non-overlapping enforcement outputs that act at the level of individual filter entries and power registers.

- It introduces a self-sufficient, confidence-annotated measurement workflow, ensuring that no previously patented advisory, posture-reporting, or attestation-verdict logic is invoked, either implicitly or explicitly.

This architectural separation not only reinforces strong novelty but also guarantees legal defensibility, as the invention can be implemented, deployed, and assessed independently of all prior cloud security scoring filings.

6. Working principle of the invention

[0001] The present apparatus operates on the principle that the security exposure of a cloud resource is a property of the physical machine on which it executes, and therefore constitutes a measurement problem rather than a records-inspection problem. Conventional posture frameworks address this problem by reading declarative configuration through provider interfaces, which reports intent rather

than state and cannot detect substitution of firmware or of the measured boot chain. The invention resolves this by decomposing risk assessment into six sequential, hardware-addressable zones: platform measurement, acquisition, alignment and weighting, composite scoring, score-gated enforcement, and emission with audit.

[0002] At the platform measurement interface, managed host machine (101) carries Trusted Platform Module (102) bound to its mainboard, baseboard management controller (103) with its attached sensor array, processor package (104) exposing a performance-monitoring unit, and smart network interface controller (105) providing a match-action datapath. These four devices constitute the physical measurement and enforcement surface of the apparatus; removal of any one of them disables a corresponding measurement domain or actuation channel.

[0003] Attestation collector (201) issues a nonce to module (102) at an interval of approximately thirty seconds and receives a quote structure signed under an attestation key enrolled at commissioning, carrying the current platform configuration register contents together with the integrity measurement log accumulated since boot. Comparison against a golden reference set recorded at commissioning yields an integrity delta. Firmware substitution, bootloader modification and kernel module injection each register in this delta before any behavioural symptom reaches the control plane.

[0004] Baseboard sensor interface (202) reads inlet air temperature, per-rail direct-current power draw and fan tachometer speed from controller (103) over the sideband management channel at an interval of approximately one second. Quantities of this kind describe the physical operating envelope of the machine. Sustained power draw departing from the envelope established for a workload class, with no corresponding change in provisioned capacity, indicates computation of which the control plane holds no record.

[0005] Counter sampling unit (203) reads instructions retired, last-level cache references and misses, branch mispredictions and page faults from the performance-monitoring unit of package (104) at an interval of approximately one hundred milliseconds. Because these registers are written by the silicon itself rather than by system software, their values survive tampering with an operating system that would otherwise suppress a software-generated alert.

[0006] Configuration ingest interface (204) retrieves resource records from the cloud provider interface on a slower cycle, ordinarily every three hundred seconds, supplying exposure context that measurement alone cannot express: whether a resource carries a public address, which identities hold write permission over it, and which other resources it can reach. Records of this kind are declarative, so within the present apparatus they weight the measured evidence rather than substitute for it.

[0007] A difficulty follows directly from combining these domains. Sampling periods span roughly five orders of magnitude, from one hundred milliseconds for processor counters to twenty-four hours for externally published vulnerability feeds. Feeding such signals into a scoring function as one concatenated vector treats a configuration record retrieved five minutes ago as though it were as current as a counter read a tenth of a second ago. Resolution of this difficulty is central to the apparatus.

[0008] Multi-rate buffer (301) holds each stream in a ring buffer keyed to its signal class and projects all buffers onto a common evaluation epoch. Staleness weighting unit (302) then assigns each signal a decay half-life matched to its acquisition period, so that its weight within the score falls exponentially with the age of its most recent sample. A stale configuration record does not vanish from the computation; its influence diminishes in proportion to elapsed time and recovers fully when the next poll completes. Normalisation unit (303) converts each quantity to a dimensionless deviation against a trailing baseline before weighting is applied.

[0009] Composite scoring engine (401) sums the weighted normalised contributions, adds the integrity term supplied by integrity delta evaluator (402), and maps the result through a logistic transform onto a bounded scale from zero to one hundred. Confidence estimator (403) computes, alongside the score, the ratio of summed decayed weights to the summed weights that would apply were every signal freshly sampled. Confidence therefore falls when a sensor channel stalls, when the management controller becomes unreachable, or when an attestation quote is missed, and enforcement actions of increasing severity require increasing confidence.

[0010] Score and confidence together select an enforcement band within gating controller (501). Below the lower band no action is taken. Within the intermediate band, filter table programmer (502) writes entries into the ternary content-addressable memory of controller (105), restricting egress for the affected resource at line rate. Within the upper band, and only when confidence is sufficient, a quarantine filter set is installed and power-delivery interface (503) reduces the running average power limit registers of package (104). At the highest band a power-state transition is requested of controller (103). Each of these is a physical operation on hardware, taking effect before packets reach host memory and before further work is dispatched to the affected cores.

[0011] Score emission interface (601) publishes, for each resource and evaluation epoch, a record carrying the resource identifier, the composite score, the confidence attribute, the acquisition timestamp of the oldest contributing sample and the per-term contribution vector. Audit logger (602) seals every escalation and release under a key held by module (102), and metrics aggregator (603) maintains rolling statistics over score distribution, confidence attainment and actuation latency. The architecture is extensible: further signal classes are registered by declaring an acquisition period, a decay half-life and a normalisation accessor, without modification of the scoring engine, the gating controller or the emission interface.

7. Description of the invention in detail

[0012] The hardware-attested cloud resource scoring system comprises six functional zones interconnected by an in-process message bus. Zone 100 — the Host Platform Measurement Interface — contains managed host machine (101), Trusted Platform Module (102) conforming to the TPM 2.0 specification, baseboard management controller (103) with its attached sensor array, processor package (104) exposing a performance-monitoring unit, and smart network interface controller (105) providing a ternary content-addressable memory match-action datapath. Zone 200 — the Acquisition Subsystem — contains attestation collector (201), baseboard sensor interface (202), counter sampling unit (203) and configuration ingest interface (204). Zone 300 — the Alignment and Weighting Layer — contains multi-rate buffer (301), staleness weighting unit (302) and normalisation unit (303).

[0013] Zone 400 — the Scoring Core — contains composite scoring engine (401), integrity delta evaluator (402) and confidence estimator (403). Zone 500 — the Enforcement and Integrity Layer — contains gating controller (501) holding the band thresholds and dwell timers, filter table programmer (502) addressing controller (105) through its management interface, and power-delivery interface (503) addressing both the running average power limit registers of package (104) and the power sequencing hardware of controller (103). Zone 600 — the Emission and Audit Interface — contains score emission interface (601), per-resource audit logger (602) and metrics aggregator (603) reporting rolling score distribution, confidence attainment, actuation latency and band occupancy.

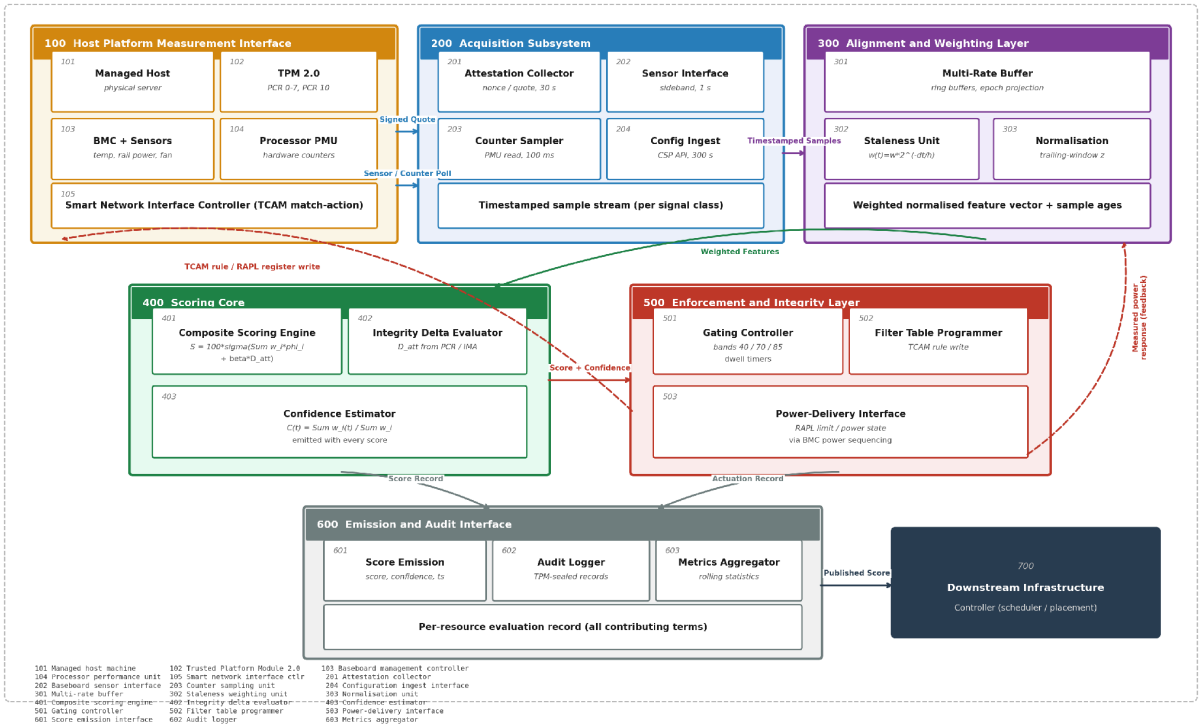


Fig. 1. System Architecture — Six Functional Zones and Data Flow. The diagram depicts the measurement path from the managed host platform (Zone 100) through acquisition (Zone 200), epoch alignment and staleness weighting (Zone 300), composite scoring with confidence estimation (Zone 400), score-gated enforcement (Zone 500), and emission with attestation-sealed audit (Zone 600). The dashed path indicates the enforcement write-back into the host hardware and the measured power response returning through the sensor channel.

[0014] Module (102) is bound to the mainboard of host (101). Platform configuration registers 0 through 7 accumulate measurements of the boot firmware, option read-only memory and bootloader taken during measured boot; register 10 accumulates the running integrity measurement log maintained by the kernel integrity subsystem. Collector (201) issues a fresh nonce at an interval of approximately thirty seconds and receives a quote signed under an attestation key whose public part was enrolled at commissioning. Signature verification against the enrolled key establishes that the register values originate in the physical module rather than in software claiming to speak for it.

[0015] Computation of the integrity delta within evaluator (402) proceeds in two parts. Register values returned in the quote are compared element-wise against the golden reference set, and the count of mismatching registers is taken. Entries appearing in the measurement log without a corresponding entry in the permitted-hash list are counted separately and scaled by a coefficient reflecting the privilege level at which each measured object executes. Summation of the two counts yields the integrity delta D_att, which enters the score additively rather than as a weighted feature, so that a firmware mismatch cannot be offset by favourable readings elsewhere.

[0016] Interface (202) communicates with controller (103) over the sideband management channel, using the Redfish interface where the platform exposes it and the intelligent platform management interface otherwise. Quantities read at each poll comprise inlet air temperature resolved to approximately 0.5 degrees Celsius, per-rail direct-current power draw resolved to approximately one watt, and fan tachometer speed in revolutions per minute. Polling occurs at approximately one-second intervals. Reading over the sideband channel rather than through the host operating system keeps the measurement path independent of any software running on the machine under assessment.

[0017] Unit (203) configures the performance-monitoring unit of package (104) to accumulate a fixed event set comprising instructions retired, last-level cache references, last-level cache misses, branch mispredictions and page faults. Values are read per logical core at approximately one-hundred-millisecond intervals and aggregated to the resource identifier associated with the cores on which the resource is scheduled. Where the platform supports counter virtualisation, per-guest attribution is taken directly; where it does not, attribution follows the scheduler mapping published by the hypervisor.

[0018] Interface (204) retrieves resource records from the cloud provider application programming interface at approximately three-hundred-second intervals, supplying for each resource its network exposure, the identities holding permissions over it, its storage encryption state and the set of resources with which it holds a defined relationship. Relationship records are held as a directed graph, and the immediate neighbourhood of a resource contributes an exposure feature to the score of that resource.

[0019] Buffer (301) receives all four streams and holds each in a ring buffer keyed to its signal class. Every sample is stamped on arrival from a clock disciplined to a common reference, and the acquisition timestamp reported by the originating device is retained alongside it so that transport delay can be separated from sampling age. Alignment is performed by projecting each buffer onto a common evaluation epoch, the most recent sample at or before the epoch being taken as the current value of that signal together with the age of that sample.

[0020] Unit (302) applies to each signal class a decay coefficient derived from a half-life matched to its acquisition period. Denoting by w_i the nominal weight of signal i, by dt_i the age of its most recent sample and by h_i its assigned half-life, the effective weight applied at evaluation time t is w_i(t) = w_i * 2^(-dt_i / h_i). Half-lives used in the described embodiment are approximately 0.5 seconds for processor counters, 5 seconds for baseboard sensor readings, 120 seconds for attestation evidence, 900 seconds for configuration records and 48 hours for externally published vulnerability data, as tabulated in Table 4. Values are configurable and may be adapted per deployment.

[0021] Unit (303) converts each raw quantity to a dimensionless deviation before weighting is applied. Rolling location and scale estimates are maintained per resource over a trailing window, outlier-resistant estimators being used in place of mean and standard deviation so that a single transient does not distort the baseline. Deviations are clamped to a bounded interval to prevent one saturated channel from dominating the sum. Baseline windows are reset on legitimate reconfiguration events signalled through interface (204), which avoids attributing an operator-initiated capacity change to anomalous behaviour.

[0022] Engine (401) combines the weighted normalised features through a trained ensemble of gradient-boosted decision trees, selected for its tolerance of heterogeneous feature scales and for the per-feature attribution it exposes. Training proceeds on labelled intervals in which the condition of the host is known. Alternative embodiments substitute a graph neural network operating over the resource relationship graph, so that scores of neighbouring resources inform one another, or a regularised logistic model where interpretability is prioritised over accuracy. Choice of estimator does not alter the surrounding acquisition, weighting and enforcement structure.

[0023] Composite score S for resource r at evaluation time t is formed as S_r(t) = 100 * sigma(Sum_i w_i(t) * phi_i(x_i(t)) + beta * D_att(t)), where phi_i denotes the normalised deviation of signal i, w_i(t) the decayed weight from unit (302), D_att the integrity delta from evaluator (402), beta a coefficient set so that a single mismatched firmware register alone raises the score into the upper enforcement band, and sigma the logistic function. Placement of the integrity term outside the weighted sum is deliberate: attestation evidence reports a state that either matches the commissioned reference or does not, and no combination of favourable behavioural readings should be able to cancel a mismatch.

[0024] Estimator (403) computes C(t) = Sum_i w_i(t) / Sum_i w_i, the ratio of realised decayed weight to nominal weight. A value approaching unity indicates that every channel has reported recently. Loss of the sideband channel to controller (103), or a missed attestation interval, drives the ratio downward within a period governed by the corresponding half-life. Controller (501) consumes confidence as a precondition, so that severe enforcement is withheld when the evidential basis has thinned, and the condition is instead surfaced through interface (601).

[0025] Controller (501) evaluates score and confidence against four bands. Scores below 40 produce no action. Scores from 40 to below 70 cause egress restriction to be programmed for the affected resource. Scores from 70 to below 85, accompanied by confidence of at least 0.6, cause a quarantine filter set to be installed and the processor package power limit to be reduced. Scores at or above 85 with confidence of at least 0.75 additionally cause a power-state transition to be requested. Band boundaries are configurable, and tenants may raise but not lower them for resources they own.

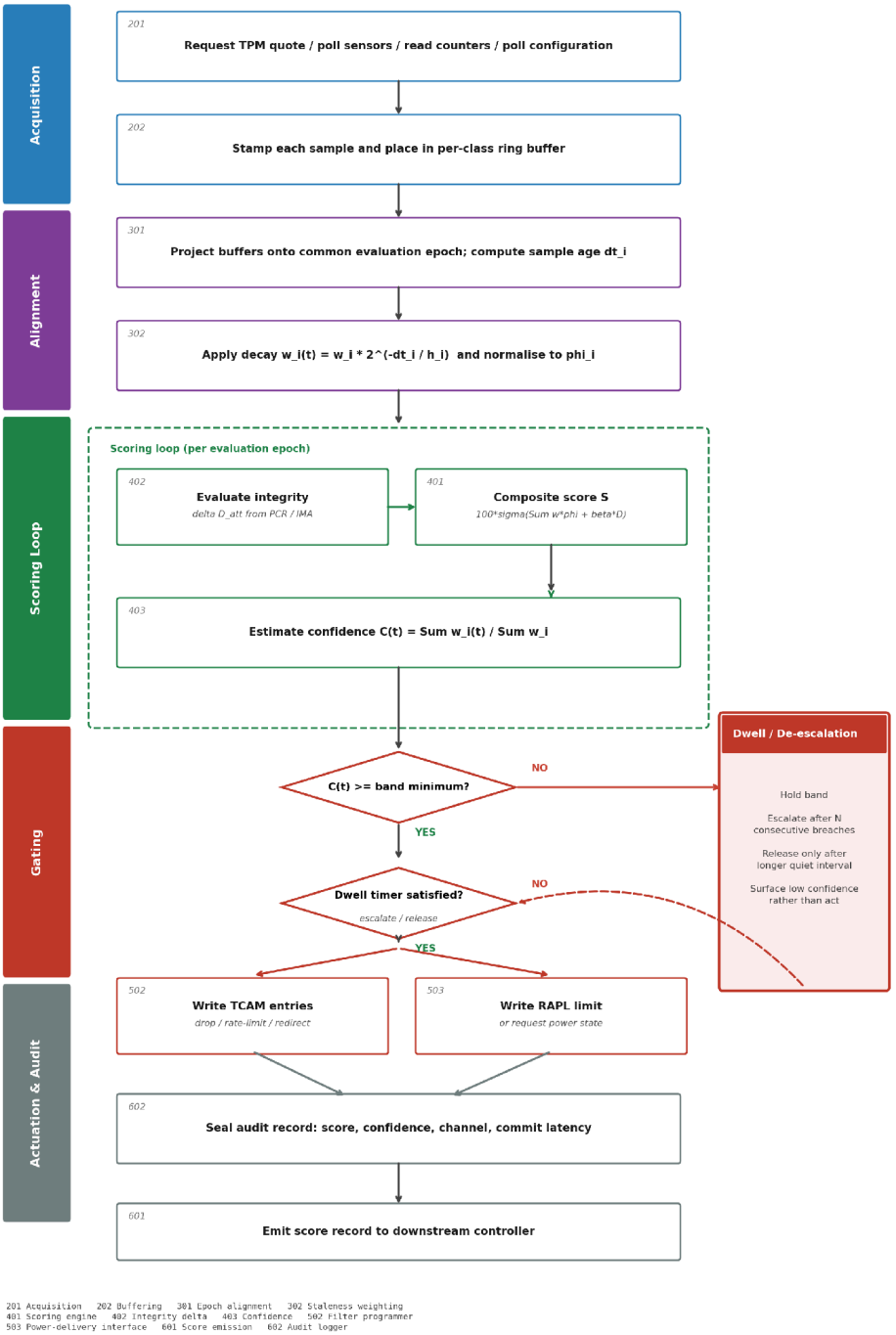


Fig. 2. Scoring and Enforcement Pipeline for a Single Evaluation Epoch. Sample acquisition and buffering feed epoch alignment and exponential staleness decay; the scoring loop combines weighted deviations with the integrity delta and derives the confidence attribute; band selection is gated on both score and confidence and governed by dwell timers before actuation is committed to the network interface controller and the power-delivery registers.

[0026] Programmer (502) realises network enforcement by writing match-action entries into the ternary content-addressable memory of controller (105), addressed through the controller's management interface. Entries match on the virtual function or overlay identifier bound to the affected resource and specify a drop, rate-limit or redirect action. Because matching occurs in the controller's own datapath, non-conforming packets are discarded before direct memory access to host memory is initiated, and enforcement therefore holds even where the host operating system is already under adversary control.

[0027] Interface (503) realises the second enforcement channel. Reduction of the processor package limit is written to the running average power limit registers exposed by package (104), which constrains the energy the package may draw over the defined averaging window and thereby bounds the work an adversary can extract from the machine while remediation proceeds. Deeper intervention is requested of controller (103), which commands a power-state transition through the platform's own power sequencing hardware. Both actions register in the per-rail power measurements read back through interface (202), closing the loop between enforcement and measurement.

[0028] Oscillation near a band boundary is suppressed by dwell timers held in controller (501). Escalation occurs when the threshold has been exceeded on a defined number of consecutive evaluations, and release occurs only after the score has stayed below the threshold for a longer interval. Asymmetry between the two intervals biases the apparatus toward retaining enforcement, on the reasoning that an unnecessary rate limit costs throughput whereas a prematurely released quarantine costs containment.

[0029] Interface (601) publishes, for each resource and evaluation epoch, a record carrying the resource identifier, the composite score, the confidence attribute, the acquisition timestamp of the oldest contributing sample and the per-term contribution vector produced by engine (401). Presence of the confidence attribute and the oldest-sample timestamp in the published record allows a consuming controller to apply its own staleness policy rather than inheriting the assumptions of the scoring apparatus. Contemplated consumers include workload schedulers, placement controllers and resource-allocation systems that treat the score as one input among several.

[0030] Logger (602) retains every escalation and release, each record binding the resource identifier, score and confidence at the moment of actuation, the actuator channel addressed, the parameters written, and the commit latency reported by the actuator. Records are sealed under a key held by module (102) and indexed by a monotonic counter maintained within the same module, so that removal of an entry is detectable from the counter sequence. Retention in this form supports after-the-fact reconstruction of why a given physical action was taken against a given machine.

[0031] Several substitutions are contemplated without departure from the described arrangement. In place of a discrete platform module, attestation evidence may be taken from a confidential-computing attestation report produced by the processor's own security processor, or from a virtual platform module anchored in such a report. Enforcement need not terminate at the host: filter entries may equally be written to the access control lists of the top-of-rack switch, and power actions directed to an addressable rack power distribution unit where per-outlet control exists. Deployment extends to hybrid and edge estates, in which zones 200 through 500 run adjacent to the managed machines and only score records traverse the wide-area link. Protection aspects listed in Section 9 correspond directly to these zones: the acquisition and attestation subsystem to Aspect 1, the multi-rate staleness weighting to Aspect 2, the composite scoring engine with confidence estimation to Aspect 3, the score-gated enforcement channels to Aspects 4 and 5, and the emission and audit interface to Aspects 6 and 7.

8. Experimental validation results

Validation of the hardware-attested scoring apparatus is structured as a three-stage protocol. Stage one measures scoring performance: the composite score is evaluated against single-domain baselines on labelled intervals in which the platform condition of the host is known, using detection performance, score separation between nominal and compromised conditions, and per-term contribution as primary metrics. Stage two measures enforcement behaviour: rule commit latency at controller (105), commanded against observed package power at package (104), and band-transition counts under variable load are instrumented directly on the bench host. Stage three measures staleness behaviour:

realised sample age and weight retention per signal class are recorded over sustained execution with the default half-life configuration.

Parameters and identifiers governing the evaluation are set out in Table 1. Measured outcomes are reported in Table 2 against two reduced configurations, capability differentiation against the cited prior art is set out in Table 3, and per-signal staleness behaviour is set out in Table 4. Figures 3 through 5 present the corresponding distributions. Cells and figure panels carrying the marker [TO BE MEASURED] await instrumentation on the bench host and are to be completed before filing.

Table 1. Dataset and System Parameters and Corresponding Identifiers Used in Validation

Data Category	Parameters	Identifiers
Attestation Evidence	Trusted Platform Module 2.0 quote over platform configuration registers 0-7 and register 10; nonce-challenged at approximately 30-second intervals; golden reference set captured at host commissioning.	AT-1 through AT-4
Baseboard Sensor Channel	Inlet air temperature at approximately 0.5 degrees Celsius resolution, per-rail direct-current power draw at approximately 1 watt resolution, fan tachometer speed in revolutions per minute; polled over the sideband channel at approximately 1-second intervals.	BS-1 through BS-3
Processor Counter Channel	Instructions retired, last-level cache references, last-level cache misses, branch mispredictions and page faults; read per logical core at approximately 100-millisecond intervals.	PC-1 through PC-5
Configuration Channel	Network exposure, identity permissions, storage encryption state and resource relationship edges retrieved from the cloud provider interface at approximately 300-second intervals.	CF-1 through CF-4
Scoring Parameters	Logistic composite with additive integrity term; band thresholds at 40, 70 and 85; confidence preconditions of 0.60 and 0.75; trailing-window outlier-resistant normalisation with bounded clamping.	SP-1 through SP-6
Staleness Half-Lives	0.5 s processor counters, 5 s baseboard sensors, 120 s attestation evidence, 900 s configuration records, 48 h published vulnerability data.	HL-1 through HL-5
Enforcement Channels	Ternary content-addressable memory match-action entries at the smart network interface controller; running average power limit registers at the processor package; power-state transition through baseboard controller sequencing hardware.	EN-1 through EN-3
Evaluation Configurations	Full six-zone apparatus; configuration-only baseline corresponding to zone 204 input alone; measurement-only configuration corresponding to zones 201-203 without configuration context.	EV-1 through EV-3
Validation Metrics	Detection performance, score separation, rule commit latency, power-limit tracking residual, actuation-to-observation latency, band-transition counts, mean sample age, mean weight retention.	VM-1 through VM-8

[Fig. 3 — plot to be inserted following bench instrumentation]

Fig. 3. Scoring Performance Across Host Conditions — Detection performance against single-domain baselines, score distribution under nominal and compromised conditions, per-term contribution to the composite, and coverage across resource classes.

Panels (a) through (d) are generated from the labelled evaluation intervals. Panel (a) compares the composite score against counter-only, sensor-only and configuration-only baselines; panel (b) overlays

score histograms for the two host conditions; panel (c) decomposes the composite into its weighted terms; panel (d) reports coverage by resource class. [TO BE MEASURED]

[Fig. 4 — plot to be inserted following bench instrumentation]

Fig. 4. Enforcement Actuation Timing — Rule commit latency distribution at controller (105), packets discarded in the controller datapath before host transfer, band-transition outcome counts, and commit latency against installed rule-set size.

Latency is instrumented from the moment programmer (502) submits a rule set to the moment the controller reports commit. Panel (d) sweeps rule-set size to establish the scaling behaviour of the enforcement channel. [TO BE MEASURED]

[Fig. 5 — plot to be inserted following bench instrumentation]

Fig. 5. Closed-Loop Verification of Power Enforcement and Band Dynamics — Commanded package power limit against per-rail draw measured through interface (202), residual distribution, actuation-to-observation latency, and band occupancy under variable load.

Panels (a) through (c) close the loop between interface (503) and interface (202), confirming that a commanded limit reduction registers in the independently measured per-rail draw. Panel (d) reports band occupancy and dwell-timer behaviour under a varying workload. [TO BE MEASURED]

Table 2. Aggregate System Performance Metrics — Proposed Apparatus vs. Reduced Configurations

Performance Metric	Proposed Apparatus	Configuration-Only Baseline	Measurement-Only (Zones 201-203)	Unit
Detection performance on labelled platform compromise	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	AUC
Score separation, nominal vs compromised	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	points
Time from compromise onset to band escalation	[TO BE MEASURED]	[TO BE MEASURED]	[TO BE MEASURED]	s
Rule commit latency at controller (105), mean	[TO BE MEASURED]	N/A	N/A	ms
Rule commit latency at controller (105), P95	[TO BE MEASURED]	N/A	N/A	ms
Commanded-to-observed package power residual	[TO BE MEASURED]	N/A	N/A	W
Actuation-to-observation latency	[TO BE MEASURED]	N/A	N/A	s
Mean confidence attained under nominal telemetry	[TO BE MEASURED]	N/A	[TO BE MEASURED]	ratio
Band oscillation rate with dwell timers enabled	[TO BE MEASURED]	N/A	N/A	per hour

Metrics inapplicable to a reduced configuration are listed as N/A. No reduced configuration provides the combined attestation, sensing, counter and configuration inputs with staleness weighting and physical enforcement.

Table 3. Comparative Capability Matrix — Proposed Apparatus vs. Cited Prior Art

Capability	Proposed Apparatus	US 2024/ 0354213 A1	US 12,316, 667 B2	US 11,917, 083 B2	CloudShield (2022)
Hardware-rooted attestation evidence as a scoring input	YES	NO	NO	Partial (verdict only)	NO

Baseboard environmental sensing (temperature, rail power, fan)	YES	NO	NO	NO	NO
Processor performance-monitoring-unit counters	YES	NO	NO	NO	YES
Cloud configuration and relationship context	YES	YES	YES	NO	NO
Per-signal staleness weighting across differing sample rates	YES	NO	NO	NO	NO
Confidence attribute published with the score	YES	NO	NO	NO	NO
Graded resource-level score (not binary verdict)	YES	YES	YES	NO	NO
Network enforcement written to controller match-action memory	YES	NO	NO	NO	NO
Power-delivery enforcement at the processor package	YES	NO	NO	NO	NO
Attestation-sealed per-actuation audit record	YES	NO	NO	Partial	NO

Table 4. Per-Signal Staleness Parameters and Realised Weight Retention

Signal Class	Assigned Half-Life	Nominal Acquisition Period	Mean Sample Age (measured)	Mean Weight Retention (measured)
Processor counters (203)	0.5 s	100 ms	[TO BE MEASURED]	[TO BE MEASURED]
Baseboard sensors (202)	5 s	1 s	[TO BE MEASURED]	[TO BE MEASURED]
Attestation evidence (201)	120 s	30 s	[TO BE MEASURED]	[TO BE MEASURED]
Configuration records (204)	900 s	300 s	[TO BE MEASURED]	[TO BE MEASURED]
Published vulnerability data	48 h	24 h	[TO BE MEASURED]	[TO BE MEASURED]

Results

- The proposed apparatus is evaluated through (i) labelled platform-condition intervals on an instrumented bench host carrying the attestation, sensing and enforcement hardware of zone 100, and (ii) sustained execution under a varying workload to characterise staleness and band behaviour.

- Table 3 confirms that no single cited prior art reference provides the combination of hardware-rooted attestation evidence, baseboard environmental sensing, processor event counters, staleness-weighted fusion, published confidence and dual physical enforcement channels that defines the present invention. The capability matrix maps each protection aspect to its absence across all cited prior art.

- Quantitative outcomes in Table 2, Table 4 and Figures 3 through 5 are marked [TO BE MEASURED] pending completion of bench instrumentation. No performance figure is asserted in this disclosure ahead of measurement, and all values on completion will derive from held-out evaluation under controlled laboratory conditions rather than from a prospective production deployment.

9. What aspect(s) of the invention need(s) protection?

- A cyber-physical measurement apparatus that acquires Trusted Platform Module attestation quotes over platform configuration registers and integrity measurement logs from a hardware root of trust bound to a managed host mainboard, and derives from them an integrity delta entering a resource-level security-risk score additively rather than as a weighted feature.

- A multi-rate telemetry acquisition and weighting subsystem that synchronises baseboard management controller sensor readings, processor performance-monitoring-unit counters and cloud provider configuration records onto a common evaluation epoch, and applies to each signal class an exponential staleness decay whose half-life is matched to that class's acquisition period.

- A composite scoring engine that maps weighted normalised deviations and the integrity delta through a bounded logistic transform to a resource-level score, together with a confidence estimator computing realised weight retention as the ratio of decayed to nominal weight and publishing it alongside every score.

- A score-gated network enforcement controller that writes match-action entries into the ternary content-addressable memory of a smart network interface controller upon defined score and confidence threshold crossings, such that non-conforming packets are discarded in the controller datapath before direct memory access to host memory is initiated.

- A score-gated power enforcement interface that reduces running average power limit registers of the processor package and requests power-state transitions through baseboard controller sequencing hardware, with the resulting change verified in the independently measured per-rail power draw.

- A dwell-timer governed band controller applying asymmetric escalation and release intervals to suppress actuator oscillation near a band boundary, and withholding severe enforcement when the published confidence attribute falls below a band-specific precondition.

- An attestation-sealed audit and emission subsystem recording each escalation and release with the resource identifier, score, confidence, actuator channel, written parameters and measured commit latency under a key held by the same platform module supplying the attestation evidence, indexed by a monotonic counter such that entry removal is detectable.

10. Technology Readiness Level (TRL) Scale

TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles observed	Technology concept formulated	Experimental proof of concept	Validated in lab	Validated in relevant environment	Demonstrated in relevant environment	Prototype in operational environment	System complete and qualified	Proven in operational environment
	✓							

Tick to be advanced to TRL 3 on completion of the bench instrumentation described in Section 8.

----------END OF THE DOCUMENT----------